SYED ANWAR

Sr. AI/ML Engineer

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Islamabad, Pakistan

SUMMARY

Senior Data Scientist & AI/ML Engineer with 6+ years of experience designing, training, and deploying production Machine Learning, Deep Learning, and Generative AI systems. Specialized in Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Computer Vision, and multi-agent AI architectures — from training Transformer models from scratch to fine-tuning LLMs and shipping systems that scale. Delivered 50+ AI/ML projects for clients across the USA, UK, Middle East, and Europe

KEY ACHIEVEMENTS

Client Projects Completion

Completed over 50 AI/ML projects for clients with a 98% satisfaction rate.

Transformer Model Creation

Developed 5M parameter text generator, achieving 74% accuracy.

AI Receptionist Development

Built AI receptionist system, reducing response times to 850ms.

Attendance System Automation

Automated facial recognition attendance, replacing fingerprint systems, boosting efficiency by 40%.

EXPERIENCE

07/2025 - Present

Islamabad, Pakistan

Senior AI/ML Engineer

Appedology Pvt Ltd

- Devised an AI-powered Medical Coding platform to extract clinical data from PDF/image-based documents and generate structured ICD-10, CPT, HCPCS, and modifier codes.
- Built specialty-driven AI workflows with OCR extraction, ICD-10 hierarchy lookup, multi-document validation, and JSON-based code generation outputs.
- Executed bulk CSV/XLSX code generation and audit agent workflows, secure JWT admin APIs, user approval management, and usage-based billing logic.
- Integrated Stripe payment gateway for real-time transactions and applied cost limitation strategies.

02/2025 - 07/2025

Islamabad, Pakistan

Lead AI/ML Engineer

AlphaSquad

- Built and fine-tuned a persona-driven crypto Twitter agent (Mind of Pepe) by scraping and cleaning tweets from the @aixbt agent and adapting a Llama model to mimic its style, achieving 98% signal accuracy and viral engagement
- Shipped a gated terminal integrating AI-based signal generation, crypto sentiment analysis, and technical analysis using real-time CoinMarketCap data, delivering actionable insights to meme coin traders
- Integrated a Retrieval-Augmented Generation (RAG) system with the agent and terminal, enabling contextual understanding by pulling live crypto data and user behavior patterns
- Built a full AI-powered receptionist system using n8n, Telnyx, and ElevenLabs for Jumper Media, enabling intelligent call handling, event scheduling, follow-ups, and outbound calls
- Integrated real-time speech-to-text transcription and RAG with webhook-driven context from customer emails and call logs, improving personalization and relevance
- Optimized system performance, reducing response latency to 850ms for a seamless, human-like user experience

08/2024 - 02/2025

Lahore, Pakistan

Data Scientist

MetaViz Pro

Delivered Computer Vision, Deep Learning, and Natural Language Processing (NLP) systems, including engineering Transformers, Large Language Models (LLMs), and AI agents from scratch; leveraged Data Science to solve business problems

- Trained two Transformer-based models from scratch using GPT-4o and PyTorch: a 5M-parameter text generation model (74% accuracy) and a 19M-parameter Python/Java code completion model (67% accuracy)
- Built a real-time yoga pose estimation system using computer vision, comparing extracted pose features against a reference database and providing corrective feedback
- Engineered a facial recognition attendance system with PPE compliance monitoring, triggering alerts and identifying non-compliant workers in real time
- Optimized a Convolutional Neural Network (CNN) for brain tumor detection, achieving 98% accuracy with a 5-layer architecture — matching EfficientNet-level performance with a far more compact model
- Built an AI agent system in LangGraph to score AI job-replacement risk, perform skills gap analysis, and recommend career pathways
- Managed and cleaned complex financial data, engineered machine learning models for insights, and analyzed trends to support strategic financial decisions

EXPERIENCE

09/2023 - 08/2024

Peshawar, Pakistan

Data Scientist

JMM Technologies

Created Computer Vision, Deep Learning, and NLP chatbot systems using Large Language Models (LLMs); employed Data Science to solve business problems

- Engineered an AI web app using GPT-4o to detect translation errors across Spanish, French, and Arabic documents against English source text, improving accuracy from 50% to 90% through advanced prompt engineering and reducing costs 40% via token management and Python regex
- Led a US company's end-to-end system deploying regression algorithms for football score predictions, including cleansing data from a third-party API for optimal input quality
- Built an AI system for generating custom LLM-based knowledge assistants from YouTube videos, PDFs, and website content using the OpenAI Assistant API, LangChain for video transcription, and web scraping — enabling RAG systems and code interpreters
- Fine-tuned YOLOv8 and YOLOv5 for an American Sign Language detection system (MVP stage) for Saudi Airport Assistance, achieving high detection accuracy while resolving FPS lag
- Built an advanced RAG application using Query Expansion and Cross-Encoder Re-ranking, improving retrieval accuracy
- Used BeautifulSoup, Selenium, and Octoparse for large-scale web scraping of real-time data from authorized sources
- Utilized time series modeling (ARIMA, SARIMA, Facebook Prophet) to forecast temperature and inflation trends
- Deployed ML pipelines on AWS (EC2, Lambda, S3, Cloud9) for scalable data storage and transformation
- Built sales dashboards in Power BI, turning complex datasets into actionable insights for decision-making
- Delivered data cleaning and analysis for an e-recycling company project, generating actionable business insights

11/2022 - 05/2023

Quetta, Pakistan

Machine Learning Engineer Associate

Government Innovation Lab (United Nation Development Program UNDP)

- Led a 3-person team (including 1 junior team member) in developing a facial-recognition-based Smart Attendance System achieving 94% accuracy, replacing the department's manual fingerprint-based attendance process
- Contributed to a Smart Vehicle Counting system using machine learning algorithms for precise, automated vehicle counting
- Collaborated with cross-functional teams as ICT Developer Fellow, integrating emerging technologies into government infrastructure

06/2022 - 07/2022

Quetta, Pakistan

Data Science Intern

British Airways

- Analyzed a dataset of 20,000+ booking records to identify patterns behind a 27% ticket cancellation rate, informing targeted mitigation strategies
- Designed predictive models using machine learning algorithms to optimize operational performance and forecast trends
- Translated complex data findings into actionable insights and recommendations, contributing to improved profit margins and operational efficiency

01/2021 - 12/2021

Quetta, Pakistan

Research Assistant

University of Balochistan (Computer Science Department)

- Trained and evaluated a Gender Prediction model achieving 99% accuracy under PhD supervision as part of an academic research initiative
- Implemented and compared four detection models (Faster R-CNN, R-CNN, SSD, YOLOv8) for license plate recognition, achieving up to 86% accuracy and contributing to an academic publication

EDUCATION

08/2018 - 12/2022

Quetta, Pakistan

Bachelor's in Computer Science

University Of Balochistan

CGPA **3.36** / **4.00**

SKILLS

Programming Languages

Python | **C++** | **JavaScript** | **MySQL** | **PostgreSQL**

Core Competencies

Machine Learning | **Deep Learning** | **Large Language Models (LLMs)** | **Generative AI**

Retrieval-Augmented Generation (RAG) | **Computer Vision** | **Natural Language Processing (NLP)** | **AI Agents** | **MLOps**

SKILLS

MLOps & Data Tools

MLflow | **Weights & Biases** | **DVC** | **dbt** | **Snowflake** | **Git**

ML / DL Frameworks & Libraries

TensorFlow | **PyTorch** | **Keras** | **scikit-learn** | **Ultralytics (YOLO)** | **OpenCV** | **Pillow** | **Pandas** | **NumPy**

Transformers | **LangChain** | **LangGraph** | **CrewAI** | **LlamaIndex** | **HuggingFace** | **Pydantic** | **NLTK** | **PySpark**

Web Frameworks

Django | **FastAPI** | **Flask** | **Streamlit** | **Gradio** | **Dash** | **HTML** | **CSS** | **Selenium** | **BeautifulSoup** | **Octoparse**

Cloud & DevOps

AWS (EC2, Lambda, S3, Kinesis)

Azure (OpenAI, AI Search, Document Intelligence, App Services, Container Apps, Function Apps) | **Google Cloud Platform**

IBM Cloud | **DigitalOcean** | **Docker** | **Kubernetes** | **GitHub Actions** | **CI/CD** | **Airflow**

Visualization & BI

Matplotlib | **Seaborn** | **Plotly** | **Power BI**

Hardware & IoT

NVIDIA Jetson Nano | **Raspberry Pi** | **Arduino** | **OpenCV OAK-D**

PROJECTS

Transformers From Scratch

Tech Stack: Python, GPT-NEOX, CodeXGLUE, PyTorch

- Constructed a Transformer-based text generation model (5M parameters) using GPT-4o and PyTorch, achieving 74% accuracy in generating contextually relevant text
- Created a Transformer-based code completion model for Python and Java (19M parameters), achieving 67% accuracy
- Fine-tuned and evaluated both models on task-specific datasets, achieving strong performance with compact architectures

Proof Lingo

Tech Stack: OpenAI, LangChain

- Built an AI system to detect contextual, missing, consistency, and translation errors by comparing source and target language documents using GPT-4, achieving over 90% accuracy
- Reduced system costs by 40% through token management and prompt engineering
- Created AI tools for text formatting, bilingual document queries, tone adaptation, and rephrasing; collaborated with the backend team to build RESTful APIs for CRUD operations

Sign Language Detection

Tech Stack: TensorFlow, Keras, OpenCV, Ultralytics, OAK-D

- Fine-tuned YOLOv8, LSTM, and CNN1D models for sign language detection as part of a Saudi government initiative
- Integrated the models with ChatGPT to enhance language processing capabilities
- Combined computer vision and language understanding technologies to deliver an effective sign language detection system

Smart Attendance System

Tech Stack: scikit-learn, OpenCV, NumPy, Flask, HTML, CSS, Ajax

- Built a facial recognition-based attendance system using SVM and KNN models for accurate individual recognition
- Deployed the system within the Planning and Development Department, improving operational efficiency and security

PROJECTS

Smart Vehicle Counting System

Tech Stack: OpenCV, Labeling, JetsonNano

- Collected, annotated, and cleaned vehicle image data to build a robust training dataset
- Optimized a YOLOv8 model on vehicle data, improving object detection accuracy

Self-Driving Car

Tech Stack: Raspberry Pi, Arduino, OpenCV

- Successfully executed a FYP focused on a Self-Driving Vehicle, integrating cutting-edge IoT and AI technologies
- Demonstrated proficiency in merging innovative technologies to create an advanced autonomous vehicle system
- Applied practical skills and theoretical knowledge to contribute to the forefront of self-driving vehicle development

License Plate Detection

Tech Stack: Python, TensorFlow, OpenCV, Keras, NumPy

- Implemented License Plate Detection for a research paper project under PhD supervision
- Trained four models (Faster RCNN, RCNN, SSD, Yolov8) to achieve comprehensive license plate recognition
- Contributed to advancing the field through hands-on application of diverse detection models in a real-world research setting

AWARDS

IEEE Workshop

Training program in university from IEEE. In Award, I get Certificate and also IEEE publication account for free

FIND ME ONLINE

LinkedIn

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Personal Website

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